

Mitigating Quasiparticle Induced T1 Relaxation in Transmon Qubits With Gap Engineering and Trapping Heterostructures

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Abstract

High-energy particles from background radioactivity or cosmic-ray muons impacting transmon qubits cause significant errors in operation. The energy deposited from these high-energy sources can deposit energy of order 100 keV easily into bulk material. This generates significant electron-hole pairs near the impact site that can travel a few hundred micrometers before trapping on defects within the bulk material. These are also accompanied by emission of many energetic athermal phonons, typically with energy levels well above the superconducting energy gap of the device layer. These high-energy excitations lead to fatal errors within superconducting devices that must be addressed. In this research paper I will show how utilizing quasiparticle trapping and barrier techniques can achieve a theoretical decrease of quasiparticle tunneling error rate by roughly 60% and address how combining this with gap engineering techniques could decrease this error rate even further.

1 Introduction

Superconducting quantum computers continue to be at the forefront of displaying quantum supremacy and steady progress towards practical quantum computing. Many companies such as Google, IBM, and Microsoft have all invested heavily into developing superconducting quantum computers due to their many advantages over other regimes. In 2019 Google's Sycamore-53 quantum chip displayed quantum supremacy, [1], that reduced a calculation task that would take 10,000 years on state-of-the-art supercomputers to just 200 seconds on their quantum chip. This achievement did not go without challenge as research groups, [2], have shown that this task can be performed faster on classical computers. Google's new Willow chip could prove to better solidify quantum supremacy with its new chip boasting an impressive 105 qubits and error rates as low as 0.035% for single-qubit gates, [3]. Although no quantum supremacy experiment for Willow has been published to date. Another group at the University of Science and Technology of China has developed a competing superconducting quantum computer named Zuchongzhi 3.0, [4]. Their quantum chip design is closely matched to Google's Willow chip specifications and have published a recent quantum advantage experiment. Zuchongzhi displayed an impressive computation difference of $5.9 * 10^9$ yr for the most powerful classical supercomputer compared to a few hundred seconds for Zuchongzhi 3.0. Such steady improvements to superconducting qubit architectures will make challenging quantum

supremacy ever more difficult as time progresses.

Due to the design of the Transmon qubits requiring construction layered on top of and around other bulk material and operations at a conducting materials critical temperature (T_C), [5], high energy phonon scattering can easily enter the superconducting bulk of the quantum chip or even the leads of the Josephson Junction. These phonons are generated from high-energy impacts such as gamma and muon radiation from external and internal systems. Once this high energy phonon interacts with the superconducting bulk it generates a quasiparticle (QP) and quasihole pair, [6], which diffusely scatters throughout the material, [7]. Once the QP reaches the Josephson Junction it has a high chance to tunnel through the barrier, absorbing energy from the qubit and inducing a shorter T1 relaxation time. Many methods have been proposed to reduce QP poisoning such as operating in underground facilities to protect from external radiation sources, introduce QP traps into the superconducting bulk material, and gap engineering a QP tunneling barrier to reduce QP density into the leads of the Josephson junction and mitigate the ability for QPs to absorb qubit energy to tunnel through the junction. My research proposes combining both QP trapping and gap engineering together in order to significantly reduce QP induced errors of transmon qubits and increase T1 relaxation time.

2 Quasiparticle Trapping and Barrier Dynamics

QPs are particle excitations *dressed* by a self-energy cloud due to its interactions with surrounding particles. These excitations above the Fermi surface will scatter and lose energy until they recombine or approach the lowest available QP energy state. Ignoring external interactions to the QPs, this leaves their dynamics within superconductors to be dependent on down scattering to lower energy levels or recombining between two broken pair electrons back into a cooper pair by emitting additional energy above the Fermi energy. With these QP dynamics one can design mechanisms to trap and barrier QPs across different materials [8].

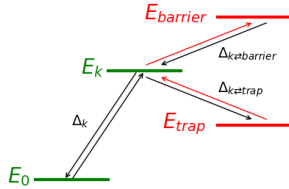


Figure 1: Simple Energy band diagram with E_0 representing the fermi energy ground state and E_k being the lowest QP excitation energy. When there is a material with a lower band gap ($\Delta_k > \Delta_{trap}$ the lowest allowed QP energy level decreases allowing for QPs to scatter into those lower energies, trapping the QP. The opposite occurs for the case of ($\Delta_k > \Delta_{barrier}$ as the QP does not have enough energy to scatter into the higher band gap material.

The lowest level QP state is dependent on material characteristics. This allows Heterostructure design where one material has a lower down scattering bounds than another connected material. By adding a lower energy band gap material island to a superconductor, QPs can scatter to one of the islands lower energy states which is not allowed in the superconductor. This *traps* the QP within the island material as shown in Figure 1. The opposite effect can also be achieved by having the second

material with a larger energy band gap than the QPs starting material. Any QPs that down scatter to or start at an energy level below the accepted energy states of the second material are unable to scatter into the material acting as a QP transport barrier. This barrier effect can be applied across a tunneling junction such as a Josephson Junction to mitigate QP tunneling [9].

3 Heterostructure Design For Quasiparticle Error Mitigation

In order to take advantage of both QP trapping and barrier dynamics, the transmon chip needs to be designed with both a QP trapping island and QP barriers to prevent QPs from entering the lead and tunneling across the Josephson Junction to limit QP tunneling. Utilizing the aluminum's properties as a superconductor and variable band gap based on thickness, a thick layer low band gap Al section and thin layer high band gap Al section can be implemented into the transmon qubit. The qubit lead will be made of the thin layer material to create a barrier preventing QPs in the larger thick layer bulk from scattering into the lead. The trap within the lead can either be the thick layer or another lower band gap energy conductive material. Any QPs not induced near the Josephson Junction will have a high likelihood of getting trapped if they manage to diffusely travel near the junction significantly reducing a large portion of QPs from tunneling. For QPs that are induced very near the junction, trapping is unlikely to reduce QP tunneling. Instead QP tunneling is mitigated through gap engineering between the leads Thin layer and the Thick layer across the junction. With a large enough band gap, most QPs will be unable to have enough energy to tunnel through the junction as they require the qubit energy ($E_q = hf_q$) and the quasiparticle energy to be greater or equal to the gap difference across the thin and thick leads of the Josephson Junction ($\delta\Delta = \Delta_{thin} - \Delta_{Thick}$). By utilizing both trapping and barrier dynamics of QP scattering, QPs not induced between the trap and the junction are significantly reduced and only high energy QPs induced near the junction will be able to tunnel over the gap.

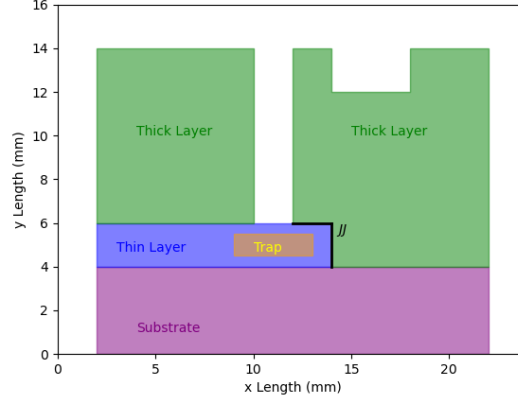


Figure 2: Heterostructure design of theoretical Transmon Qubit. Substrate represents oxidized Al as an insulating material. The Thick layer represents a thicker Al material resulting in a smaller band gap while the Thin layer represents a thinner Al material with a larger band gap ($\Delta_{thin} > \Delta_{thick}$). The total chip dimensions are 14x20 mm

4 Simulated Results

Superconducting aluminum QP scattering and recombination probabilities can be calculated with the low-frequency form approximation of the spectral weight $\alpha^2(\Omega)F(\Omega) = b\Omega^2$ [10]. Scattering and recombination lifetimes are dependent on the characteristic lifetime of QPs in Aluminum calculated to be $\tau_0 = 438$ nanoseconds (Eq. 1) with renormalization parameter $Z_1(0) = 1.43$.

$$\tau_0 = Z_1(0) \frac{\hbar}{2\pi b(k_B T_c)^3} = 438 \text{ nanoseconds} \quad (1)$$

The scattering and recombination rates can then be expressed in terms of an infinite series of modified Bessel functions K_0 and K_1 .

$$\tau_r^{-1}(\Delta(T)) = \tau_0^{-1} \left(\frac{\Delta}{k_B T_c} \right)^3 \sum_{n=1}^{\infty} \left[(-1)^{n+1} + e^{\frac{-n\Delta}{k_B T_c}} \right] \left\{ \left[4 + \frac{3k_B T}{n\Delta} + 2 \left(\frac{k_B T}{n\Delta} \right)^2 \right] K_1 \left(\frac{n\Delta}{k_B T} \right) + \left(4 + \left(\frac{k_B T}{n\Delta} \right) K_0 \left(\frac{n\Delta}{k_B T} \right) \right\} \quad (2)$$

$$\tau_s^{-1}(\Delta(T)) = \tau_0^{-1} \left(\frac{\Delta}{k_B T_c} \right)^3 \sum_{n=1}^{\infty} \left[(-1)^{n+1} + e^{\frac{n\Delta}{k_B T_c}} \right] \left\{ \left[4 - \frac{3k_B T}{n\Delta} + 2 \left(\frac{k_B T}{n\Delta} \right)^2 \right] K_1 \left(\frac{n\Delta}{k_B T} \right) - \left(4 + \left(\frac{k_B T}{n\Delta} \right) K_0 \left(\frac{n\Delta}{k_B T} \right) \right\} \quad (3)$$

At low temperatures these equations rapidly converge. The dominant terms come out to be

$$\tau_s = \tau_0 / \zeta(3.5)\Gamma(3.5) \left(\frac{T}{T_c} \right)^{7/2} \sqrt{\frac{k_B T_c}{\omega}} \quad (4)$$

$$\tau_r = \tau_0 / (\pi)^{1/2} \left(\frac{2\Delta(0)}{k_B T_c} \right)^{5/2} \left(\frac{T}{T_c} \right)^{1/2} e^{-\Delta(0)/k_B T} \quad (5)$$

QPs with energies $\omega > \Delta(T)$ have a scattering lifetime limit due to spontaneous phonon emissions. This can be calculated when $\Delta = \Delta(0)$ everywhere.

$$\tau_s = \frac{\tau_0}{\left(\frac{\Delta}{k_B T_c} \right)^3 \left(\frac{1}{3} \left[\left(\frac{\omega}{\Delta} \right)^2 - 1 \right]^{3/2} + \frac{5}{2} \left[\left(\frac{\omega}{\Delta} \right)^2 - 1 \right]^{1/2} - \frac{\Delta}{2\omega} \left[1 + 4 \left(\frac{\omega}{\Delta} \right)^2 \right] \ln \left(\frac{\omega}{\Delta} + \left[\left(\frac{\omega}{\Delta} \right)^2 - 1 \right]^{1/2} \right)} \quad (6)$$

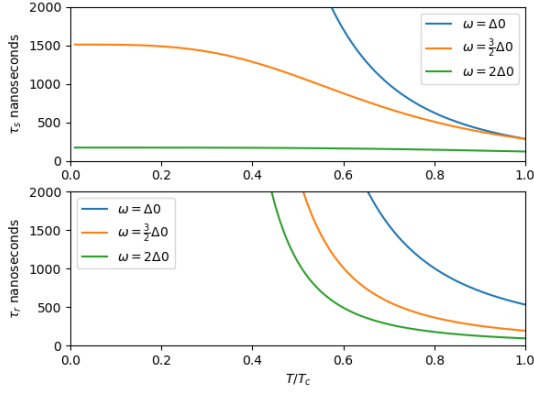


Figure 3: Above plots show the scattering and recombination lifetimes of Al dependent on the ratio of T/T_c . Scattering lifetimes with energies above the smallest allowed QP energy are limited by a constant (Eq. 6). These lifetimes that are not limited do move towards infinity as $T \rightarrow 0$

After applying the limit to scattering lifetime with $\frac{\tau_s \text{ limit} * \tau_s}{\tau_s \text{ limit} + \tau_s}$, both scattering lifetime and recombination lifetimes can be calculated with dependence to ratio $\frac{T}{T_c}$ as shown in Figure 3.

At low temperatures one can see that QP recombination lifetimes will last much longer than the duration of many gate times, showing that any QPs induced in the superconductor will only build up continuously while operating at extremely low temperatures. On the other hand, QP scattering lifetimes at energies above the lowest QP energy state will have short durations within microsecond timescales, allowing for reasonable down scattering rates for QP trapping dynamics.

To simulate the qubit chip from Figure 2, the transmon qubit is set to a temperature of $T = 20 \text{ mK}$ with a phonon injection rate of roughly ($\approx 1 \text{ phonon} / \text{nanosecond}$). To keep things simple all QPs started at $\Delta = 4\Delta_0$ and scattering probabilities into lower trap material energies were only calculated for QPs near trap bounds. The QP scattering rate can be calculated as $\frac{1}{\tau}$ and the probability for the QP to scatter to a lower energy state at a certain time (t) is $1 - e^{-\frac{t}{\tau}}$. For any QPs that reached the Josephson Junction edge it was assumed that a QP tunneling event occurred and induced T1 relaxation error. Overall QP simulation statistics were done over a period of 1 microsecond for 10 periods for both with trap and without a trap.

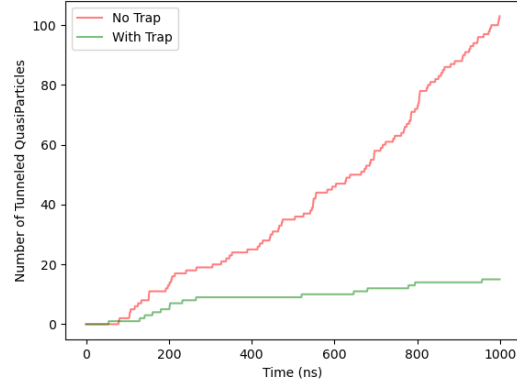


Figure 4: QP tunneling errors reported for each time interval over the 1 microsecond experimental periods. Red line represent the transmon qubit errors for no trapping designed in qubits Heterostructure while the green line includes the QP trapping material. QP tunneling errors decreased approximately 10-fold when the trapping heterostructure was introduced.

Simulating the transmon qubit chip design from figure 2 without reduced band gap Al trap resulted in a total of 991 QP tunneling errors with an average error time of 10.08 nanoseconds. With the trap resulting in only 120 total errors with an average error time of 98.63 nanoseconds. The transmon with the trap displayed nearly $\frac{1}{10}th$ of the total errors of a non-trapped Heterostructure and nearly 10-fold the time between errors. A full timeline of recorded QP tunneling errors for with and without trap transmons can be seen in Figure 4. Introducing the QP trap resulted in the error rate dropping significantly.

5 Conclusions

Designing transmon qubit heterostructures with QP trapping and barrier dynamics allow for significant reduction in QP tunneling induced errors. Under these theoretical systems this improvement can extend the qubit lifetime by a factor of 10 compared to without trapping. Due to limited time, statistics could not be gathered on how the gap engineering of the Josephson Junction decreased QP error rate further. Hopefully this can be realized with further research with better tracking of current QP energy levels of all QP particles in the system.

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